

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Parametric imitation learning via behavior cloning often loses performance when rollout errors move the agent away from the training-state distribution. We show that the same demonstrations used for training can also provide quantitative structure at inference time: a semi-parametric retrieval policy can compare each query state with nearby expert states and use those local differences to stabilize action prediction. We present **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which reparameterizes imitation learning around k -nearest-neighbor evidence rather than a single global state-to-action map. For every query, DARP retrieves expert neighbors, conditions on their actions and relative state differences, and aggregates the resulting neighbor-level action estimates. This keeps the assumptions identical to standard behavior cloning: no new data collection, no online expert feedback, and no task-specific instrumentation. Across continuous-control, robotic-manipulation, and high-dimensional visual-feature evaluations, this comparison structure yields consistent 15–46% improvements over standard behavior cloning, indicating practical gains that persist across domains, representations, and action models. Code and demos will be released upon acceptance.

1 INTRODUCTION

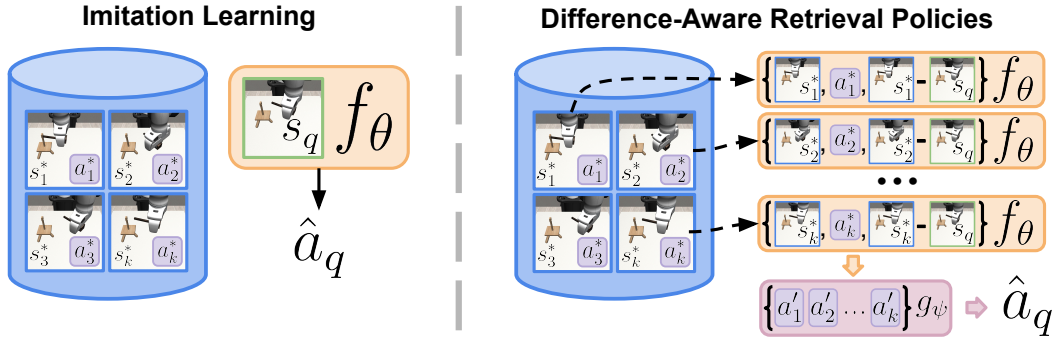


Figure 1: **Overview of DARP:** Standard BC (left) compresses the demonstrations into a parametric policy and discards the empirical state-action table at inference time. DARP (right) keeps that table active: for each query state s_q , it retrieves a local neighborhood of expert state-action pairs and predicts through difference vectors between the query and the retrieved neighbors.

Imitation learning via behavior cloning (BC) (Pomerleau, 1991) has produced strong demonstrations of robots learning complex and dexterous behaviors from expert data (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). The empirical limitation is equally clear: even when supervised losses are low, closed-loop performance can fall sharply on long-horizon tasks (Ross et al., 2011). The failure mode is *covariate shift*: small action errors compound during rollout, placing the agent in states that have little support in the demonstrations (Spencer et al., 2021; Ross et al., 2011). In those low-evidence regions, a BC policy can exhibit high-variance and unreliable actions, so modest early deviations can become task failures.

Many methods reduce compounding error by adding resources beyond the standard BC setting, such as simulators, interactive experts, large amounts of sub-optimal data, or task-specific structure (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). Our framing is deliberately stricter: we ask what quantitative gains can be obtained while using exactly the original expert state–action pairs and no additional supervision. The central question is therefore: *can the empirical geometry already present in the demonstration dataset reduce the variance of BC policies?*

Statistically, BC controls supervised risk only at expert states. That pointwise objective can fit the training distribution while leaving the policy unconstrained between and beyond observed states, precisely where closed-loop rollouts often drift. A natural way to use the data more fully is to impose *smoothness*: nearby states should produce similar actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024). Smoothness reduces spurious local fluctuations and improves rollout stability, but existing mechanisms introduce tradeoffs; augmentation does not guarantee local consistency, global priors can blur distinct behaviors, temporal penalties only compare along trajectories, and explicit graph regularizers add smoothness hyperparameters that must be selected. We return to these smoothness-based comparisons in Section 4.

The same comparison appears in the contrast between global and local learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the full demonstration set into one parametric function, which can be brittle when deployment states differ from the training distribution. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) instead consult neighborhoods of similar states and make predictions from empirical neighbors. Locality makes the evidence used for each decision explicit, but it also exposes important quantitative sensitivities: performance depends on the distance metric, naive neighborhood averaging can blur multimodal actions, and absolute-state neighbor representations can limit generalization.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a method that preserves the favorable comparison structure of local retrieval while retaining the flexibility of learned policies. At inference time, DARP (Fig. 1) retrieves k expert neighbors for the current query state, predicts one candidate action per neighbor from the neighbor state, neighbor action, and query-neighbor difference, and then aggregates those candidates with a permutation-invariant operator. This design makes every prediction conditional on concrete empirical evidence from the training set while using the learned difference model to adapt each neighbor action to the query. In spectral terms, the aggregation approximates a Laplacian filter on the k -NN (k -nearest neighbor) graph of expert states, giving a parameter-free smoothing mechanism that adapts to local data density and geometry.

We provide theoretical and empirical evidence that this evidence-aware parameterization improves BC under the same data assumptions. The analysis connects DARP to Laplacian regularization and shows that neighbor aggregation suppresses high-frequency variance while preserving low-frequency structure. Empirically, DARP improves over standard behavior cloning by 15–46% across continuous-control MuJoCo tasks, robotic manipulation in Robosuite and Robocasa, and high-dimensional visual imitation settings. The gains are not isolated to one architecture: the method extends to image features, transformers, Gaussian mixture models, and diffusion-style policies. We further include ablations that identify the quantitative sources of the margin, showing that difference vectors, meaningful retrieval, and permutation-invariant aggregation are central to the observed improvements.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

We develop an imitation-learning architecture that makes the comparison between global parametric learning and local empirical retrieval explicit. The method keeps the data requirements of standard behavior cloning, but changes how the policy uses the demonstrations: each prediction is supported by retrieved neighbors and their measured differences from the query. This section derives the theoretical basis for the resulting variance and stability gains. Readers primarily interested in the empirical comparisons may skip to Section 2.5 for a self-contained summary of DARP before proceeding to Section 3.

We first define the imitation-learning setting (Section 2.1) and then introduce a neighbor-regularized objective (Section 2.2) that exposes the statistical quantity of interest: local consistency on the demonstration manifold. Section 2.3 shows how the same smoothing effect can be obtained by

changing the policy *architecture* rather than adding a tuned regularizer. Section 2.4 then turns this construction into DARP, a semi-parametric retrieval-augmented policy that is compatible with modern neural networks and generative action models.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We use the standard finite-horizon imitation-learning setting, formalized by the Markov Decision Process (MDP) $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and P_0 is the initial state distribution. A *policy* maps states to action distributions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, to optimize task-relevant behavior; when discussing the function class, we omit the subscript θ . The only data are expert demonstrations $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$, and s^* and a^* always denote states and actions from this empirical dataset. Behavior cloning (Pomerleau, 1991) treats the comparison to expert actions as supervised learning, fitting f_θ by $\arg \max_\theta \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. The action distribution represented by f_θ can be a rich generative model (Lipman et al., 2023; Chi et al., 2024); for clarity of the initial derivation, we begin with a Gaussian parameterization¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Behavior cloning directly minimizes supervised imitation loss on expert states sampled from $\mathcal{D}^*$, but that metric does not measure how the policy changes away from the demonstrated manifold. During rollouts, accumulated errors can therefore move the agent into out-of-distribution regions where an overparameterized BC policy may produce arbitrary actions (Ross & Bagnell, 2010).

The missing evidence is the relationship among nearby demonstrations. BC compares each prediction only with the action at the same training state; it does not penalize inconsistent predictions across neighboring expert states. We therefore consider a modified objective that adds an explicit *local consistency* term: states that are close in the expert dataset should receive similar predicted actions. The resulting neighborhood-regularized loss ($\mathcal{L}_{\text{MRIL}}$) combines the standard BC supervised risk ($\mathcal{L}_{\text{BC}}$) with a smoothness penalty ($\mathcal{L}_S$), so the learned policy is evaluated both by pointwise action accuracy and by agreement with the dataset geometry.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \lambda \underbrace{\mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

Here $\ell(f(s^*), a^*)$ is the supervised imitation loss, $\mathcal{N}_k(s^*)$ is the set of k nearest expert neighbors for s^* , and $w_i(s^*)$ are normalized kernel weights determined by state differences, $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. This objective makes the quantitative comparison explicit: the supervised term measures action error at each expert state, while the smoothness term measures disagreement across local neighborhoods. As discussed below and proved in Appendix A.1.1, this is a form of *manifold regularization* or Laplacian smoothing that penalizes high-frequency variation on the expert-state graph, yielding improved variance, stability, and generalization guarantees.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

¹We show that this can be relaxed in Section 2.4

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s^{j*})\| = O(\|s^* - s^{j*}\|)$ for neighbors s^*, s^{j*} .

Thus, under the stated assumptions, MRIL has strictly stronger generalization and stability guarantees than BC under the same expert-data setting.

Proof sketch. Appendix A.1.1 gives the full proof; the sketch highlights the quantities being compared. First, the smoothness term is exactly a graph Laplacian penalty on the k -nearest-neighbor (k -NN) affinity graph with kernel weights w_i . Second, as the sample size grows, that graph penalty converges to a weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing this energy both bounds local Lipschitz variation, improving closed-loop stability, and acts as Tikhonov regularization, reducing estimator variance while keeping bias controlled. $\square$

Intuitively, the regularizer turns neighborhood agreement into a measured training criterion rather than a qualitative preference. It shrinks the local Lipschitz constant of f most strongly along high-density directions of the data distribution $p(s)$, so small state changes in well-supported regions induce small, consistent action changes. The expected practical effect is better reliability on in-distribution (ID) states and more structured extrapolation to nearby out-of-distribution (OOD) states.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although MRIL provides a clearer statistical comparison than vanilla BC, it also introduces two practical costs. The smoothness weight λ must be tuned to trade off action accuracy against local consistency, and optimizing a modified objective can change the optimization landscape. This motivates the central architectural question: *can we recover the same benefits as MRIL (Eq. 1) by changing the policy architecture while leaving the BC objective intact?*

We answer this by moving the comparison operation into the policy itself. The resulting retrieval-based architecture has an *implicit* manifold-regularization effect (iMRIL) while still training with a standard imitation objective. As illustrated in Fig. 2, the architecture supplies Laplacian-style smoothing without adding λ as a training hyperparameter. Section 2.4 builds this implicit construction into the practical DARP algorithm.

iMRIL architecture: The key change in iMRIL is to move neighborhood aggregation from the loss in Eq. 1 into the action predictor itself. Rather than train a feedforward predictor $f(s)$ with an added smoothness penalty, we parameterize the deployed predictor $\hat{f}$ as an average over predictions at retrieved training neighbors while keeping the objective as standard imitation learning. Concretely, iMRIL learns a per-state model f_θ such that the aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$ matches the expert action a^* for the query state. Thus each

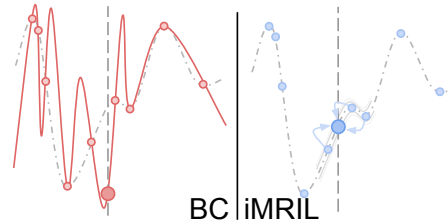


Figure 2: **iMRIL turns neighborhood evidence into Laplacian smoothing.** Compared with standard BC, which can fit jagged functions between training states, the retrieval-aggregation architecture suppresses high-variance local fluctuations and enforces consistency across the empirical graph.

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

training comparison evaluates not one isolated prediction, but the mean of k neighbor-supported estimates:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_{\theta}(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, the same comparison is used for a new state s_q : retrieve its k nearest neighbors from the training set and aggregate $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_{\theta}(s_i^*)$. Section 2.4 shows that the exact parameterization of f is empirically important; this observation leads to the difference-aware version, DARP.

This makes the variance-control mechanism visible: the supervised loss is applied after aggregation, so individual neighbor predictions cannot vary arbitrarily without affecting the aggregated action. The architecture therefore induces the same local-consistency pressure described in Section 2.2. We next state the spectral equivalence between iMRIL and Laplacian regularization.

Equivalence between iMRIL and MRIL: We defer the full formal proof relating MRIL and iMRIL to Appendix Section A.1; here we state the result and highlight the spectral comparison.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. We defer full details to Appendix Section A.1.2. The explicit regularizer admits a spectral solution by diagonalizing the k -NN Laplacian, yielding a filter of the form $(1 + \lambda \mu)^{-1}$ on each eigenmode. The implicit objective can be expressed as neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$, the random-walk matrix, which has the same eigenvectors and applies the fixed filter $1 - \mu$. Intuitively, both act as low-pass filters on the graph: modes with small eigenvalues (smooth variation across the data manifold) are largely preserved, while modes with large eigenvalues (rapid, high-variance fluctuations between neighbors) are strongly damped. Thus iMRIL implicitly performs Laplacian smoothing, reducing variance and enforcing local consistency without needing to tune λ . $\square$

The implicit-smoothing view does not remove the need to learn an accurate policy; instead, it clarifies which part of the architecture contributes which quantitative effect. The neighbor-conditioned network f_{θ} learns local action corrections, reducing bias relative to naive neighbor actions. The aggregation operator averages those proposals across the retrieved neighborhood, reducing variance and improving stability. Accuracy and stability are therefore separated into complementary components.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

We now instantiate iMRIL as a practical algorithm for large-scale imitation learning. The empirical comparisons in Section 3 depend on two concrete design choices: the representation given to the neighbor predictor, and the aggregation operator used to combine neighbor-level action candidates.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

Eq. 2 specifies aggregation but leaves the input representation of f_θ open. The practical question is what evidence each neighbor-level predictor should receive. Because the goal is to estimate how an expert action changes under a local perturbation from a neighbor to the query, DARP gives f_θ three quantities: the neighbor state s_i^* , the neighbor action a_i^* , and the difference vector $\Delta s_i = s_i^* - s_q$. Thus, rather than predict only from $f_\theta(s_i^*)$, **D**ifference-**A**ware **R**etrieval **P**olicies (**DARP**) predicts a query-specific candidate $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$ for each retrieved neighbor.

Let $\mathcal{N}_k(s_q)$ denote the indices of the k nearest expert states under a distance function $d(s_q, s_i^*)$.³ Given these retrieved neighbors, DARP computes k difference-conditioned action candidates and averages them to obtain the query action:

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

Training compares the aggregated action against the expert action for each query state in $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

This objective optimizes f_θ by the discrepancy between the aggregated prediction $\hat{a}_q$ and the expert action a_q^* . Because the loss remains a standard imitation loss, f_θ can be implemented with standard feedforward or convolutional networks. The empirical ablations in Section 3 show that the difference vector, not merely retrieval itself, accounts for a large share of the performance margin. At inference time, DARP repeats the same procedure: retrieve k neighbors and aggregate their difference-conditioned predictions as in Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

Averaging, $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, is the simplest aggregation rule and gives a transparent comparison across neighbors. It is also one member of a broader class of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. More expressive set-compliant models, such as set transformers (Lee et al., 2019) or DeepSets (Zaheer et al., 2017), can parameterize g_ψ , yielding the generalized predictor $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. This extension also broadens the action distributions that can be represented. Instead of the Gaussian distribution implicit in the L_2 objective in Eq. 5, DARP can predict parameters α of $p(a_q; \alpha)$, such as Gaussian-mixture means, covariances, and weights (Pignat & Calinon, 2019), or a diffusion score function (Chi et al., 2024). The same neighbor comparison can therefore support maximum-likelihood training for multimodal action distributions:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

At test time, the policy retrieves neighbors for s_q , constructs $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$, and samples $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$. Appendix Section A.3 gives detailed training pseudocode.

2.5 SUMMARY

In summary, local consistency in BC can be written as manifold regularization, and this Laplacian-smoothing view yields variance, smoothness, and stability improvements over standard BC (Theorem 1). Instead of adding the explicit regularizer and tuning λ , DARP builds the same smoothing pressure into the architecture through neighborhood aggregation (Theorem 2). For each expert pair (s_q^*, a_q^*) , the practical algorithm retrieves the k nearest expert states from $\mathcal{D}^*$, computes difference

³In our work we use the Euclidean distance in a pre-trained embedding space, although other neighborhood functions are also applicable. We refer the reader to Appendix Section A.2.1 for a thorough discussion of design decisions in constructing neighborhood sets via retrieval.

vectors $\Delta s_i = s_i^* - s_g^*$, maps each tuple $(s_i^*, a_i^*, \Delta s_i)$ through f_θ to produce an action candidate, and aggregates the candidates with a permutation-invariant g_ψ . The training objective remains the standard imitation objective, so the empirical comparisons isolate the effect of the retrieval-and-difference architecture rather than additional supervision.

3 EXPERIMENTAL EVALUATION

We evaluate DARP by organizing every comparison around method, baseline, metric, dataset, and evaluation condition. The experiments answer three quantitative questions: **Q1**: Does DARP consistently improve over standard behavior cloning under matched data and policy-class settings?, **Q2**: Do the gains persist for higher-dimensional state representations and multimodal action distributions?, and **Q3**: Which architectural components account for the observed margins? The evaluation spans MuJoCo continuous-control scores, Robosuite and RoboCasa manipulation success rates, real-world manipulation success rates, vision-feature policies, and discontinuous or multimodal stress tests.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) comparisons report locomotion scores for four embodiments on a flat plane: Hopper, Walker, Ant, and HalfCheetah. These tasks test whether retrieval-based local evidence improves continuous-control returns relative to BC and non-parametric retrieval baselines.

Robosuite Tasks: The Robosuite (Zhu et al., 2020) comparisons report manipulation success rates for three single-arm tasks: Stack, where the robot places a smaller cube on a larger cube; Threading, where it inserts a thin tool into a small ring; and Square Peg, where it places a square block with a central hole onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) comparisons measure success in randomized kitchen manipulation: closing an open drawer, closing a microwave door, and turning a stove knob off. These tasks test whether the BC–DARP margin remains visible under randomized scene layouts.

Baseline Comparisons: We compare DARP against baselines that isolate different sources of evidence: (1) **R&P (Sridhar et al., 2025)**: the nearest neighbor’s action, (2) **LWR (Pari et al., 2022)**: locally weighted regression over retrieved neighbors, (3) **BC**: the standard parametric behavior-cloning baseline, (4) **REGENT (Sridhar et al., 2025)**: a transformer-based in-context learner conditioned on retrieved neighbors, and (5) **MRIL**: the explicitly smoothed objective from Section 2.2. This set separates gains from retrieval alone, parametric BC alone, explicit smoothing, and the full difference-aware architecture.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

For Q1, we evaluate the core hypothesis in low-dimensional state spaces, where nearest-neighbor distances are well-defined and interpretable. The comparison spans MuJoCo locomotion scores and Robosuite/RoboCasa manipulation success rates using MimicGen-generated data (Mandlekar et al., 2023). Unless otherwise stated, g is the simple mean over the neighbor-level action predictions a' , so the reported margins reflect the difference-aware retrieval architecture rather than a learned aggregator.

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **MuJoCo low-dimensional-state comparison.** Across Hopper, Ant, Walker, and HalfCheetah, DARP gives the highest reported score in every column and improves over BC by a large margin under the same state representation. Scores are means over 100 independent trials with 95% confidence intervals. Parametric policies use Multi-Layer Perceptrons; diffusion-policy results are reported in Section 8.

The low-dimensional comparisons show a consistent pattern: DARP improves over BC in every reported environment. On robotic manipulation, the margins are 15–25 percentage points, and on MuJoCo the score improvements are substantial across all four locomotion tasks (Tables 1 and 2). Purely non-parametric retrieval methods (R&P and LWR) do not match these gains, indicating that retrieval alone is insufficient. MRIL usually improves over vanilla BC, but the highest scores in this suite are obtained by the full DARP architecture, which combines retrieval, difference conditioning, and aggregation.

These low-dimensional gains motivate the next comparison: whether the same retrieval-and-difference structure remains favorable when the state representation is high-dimensional, such as image-derived features.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations.

Given the practical instantiation in Section 2.4, we next compare DARP and BC when states are R3M image embeddings (Nair et al., 2022) rather than compact sim-

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: **Vision-based Robosuite comparison.** Success rates (%) using R3M visual features show DARP exceeding BC by 31, 38, and 35 percentage points for Stack, Thrd, and Peg, respectively.



Figure 3: RoboCasa “Drawer-Close” evaluation condition: the robot must close an open drawer in a randomized kitchen scene.



Figure 4: Robosuite “Threading” evaluation condition: the robot must insert the needle-like implement through the small ring.

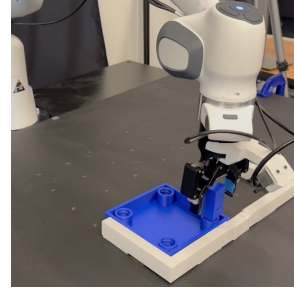


Figure 5: Real-world FurnitureBench square-table evaluation condition: the robot must pick up a table leg and screw it into a tabletop corner hole.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: **Low-dimensional robotic-manipulation comparison.** Scores are success percentages for DARP versus BC across RoboCasa, Robosuite, and the real FurnitureBench square-table task (RoboCasa and Robosuite: 100 trials; Real: 50 trials). DARP is higher in every task, with margins of 31, 16, and 15 points in RoboCasa; 25, 26, and 16 points in Robosuite; and 48 points in the real-world setting. RoboCasa and Robosuite use Multi-Layer Perceptrons, while the real-world result uses diffusion policy; see Section 8.

ulator states. Table 3 shows that DARP remains higher than BC on all three vision-based Robosuite tasks. The average improvement is approximately $\sim 35\%$, which is larger than the approximately $\sim 22\%$ average improvement observed for low-dimensional Robosuite states. Thus the empirical margin does not shrink in the visual-feature setting; it becomes larger under this evaluation condition.

Multi-modal Action Distributions. We also evaluate whether the comparison remains favorable when the expert action distribution is multimodal. On the Push-T task, DARP achieves more than a 20% improvement over behavior cloning; Appendix A.2.2 reports the task details and metric.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: To identify which comparisons produce the performance margin, we ablate the architecture on Robosuite Stack. The variants are: (1) full DARP; (2) DARP without neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with random rather than distance-based neighbors; (5) DARP using only the L_2 norm of the distance vector; (6) BC using only the query state s_q ; (7) DARP using the query state instead of the query-neighbor difference vector; and (8) DARP with a permutation-dependent, rather than permutation-invariant, aggregator over the a' values. Figure 6 reports the systematic comparison.

The ablations attribute the gain primarily to two components: the full difference vector and permutation-invariant aggregation. Removing neighbor actions has a smaller effect, whereas replacing metric-based neighbors with random neighbors sharply reduces performance, showing that the retrieval metric is part of the empirical signal. The degradation from permutation-dependent aggregation further indicates that DARP benefits from treating the retrieved set as an unordered local neighborhood.

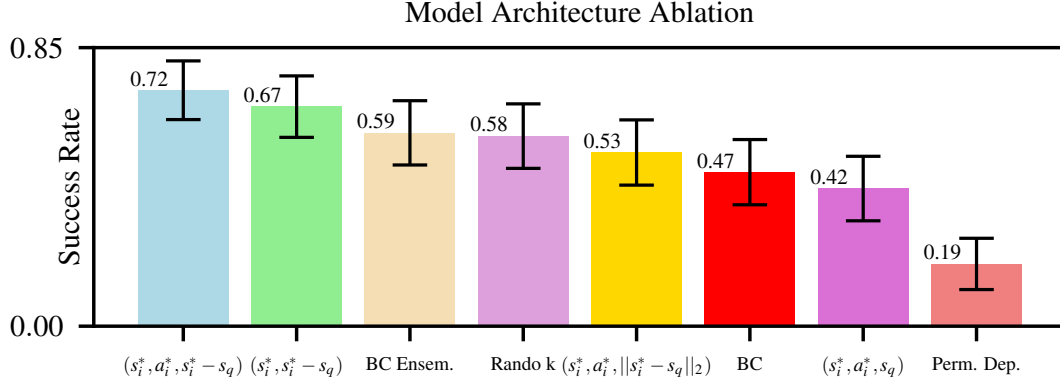


Figure 6: **Architecture ablation on Robosuite Stack.** Success rates over 100 trials with 95% confidence intervals isolate the evidence used by DARP. The largest drops occur when the difference-vector representation $(s_i^*, a_i^*, s_i^* - s_q)$ or permutation-invariant aggregation is removed, indicating that the main performance margin comes from difference-aware local comparison rather than retrieval alone.

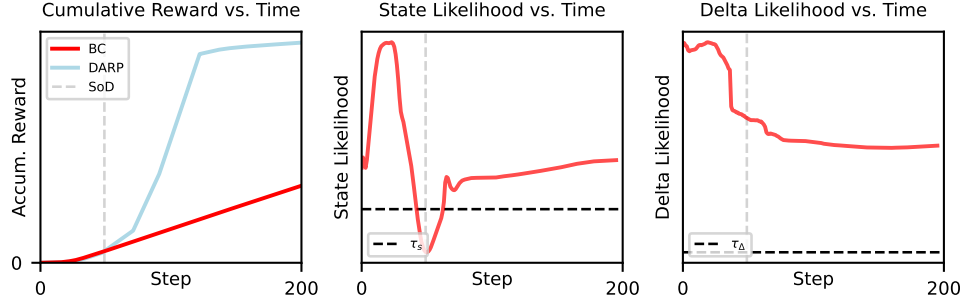


Figure 7: **Reward-divergence comparison for BC and DARP on Robosuite Stack.** The trajectories initially match, then separate at the step of divergence (“SoD”), after which BC fails and DARP succeeds. At the SoD, the query-state likelihood is $< \tau_s$ (OOD), while the query-neighbor delta likelihood is $> \tau_\Delta$ (in distribution), supporting the difference-vector framing.

Divergence Analysis: To interpret why DARP improves over BC, we compare matched rollouts where BC fails and DARP succeeds. The “step of divergence” (SoD) is the first point at which the two policies begin receiving meaningfully different reward. We set τ_s and τ_Δ to the 1st percentile of the corresponding training-set likelihoods, so values below the threshold mark the lowest-likelihood 1% of training examples. Across all six analyzed rollouts from Robosuite Stack and MuJoCo Hopper, the query state at the SoD has likelihood $< \tau_s$, while the query-neighbor delta has likelihood $> \tau_\Delta$. This consistent pattern supports the mechanism suggested by the architecture: DARP can face an OOD query state while still making an in-distribution comparison in difference-vector space. Figure 7 shows reward drift, SoDs, and likelihoods for one task, and Appendix A.2.4 provides additional robustness experiments.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL methods provide an important empirical reference point because they make local training examples visible at prediction time. VINN Pari et al. (2022) studies locally weighted regression for imitation and reports strong results in image-embedding spaces, while MiDiGaP von Hartz et al. (2025) uses mixtures of Gaussian processes to represent multimodal trajectories and support rapid generalization. SEABO Lyu et al. (2024) uses retrieval in offline RL by rewarding transitions close to retrieved neighbors, and FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) combine non-parametric retrieval with finetuning on large unlabeled datasets. DARP uses the same broad source of evidence—nearby examples—but changes the comparison made from those examples: retrieved states are represented as $(s_i^*, a_i^*, s_i^* - s_q)$ tuples and passed through a learned semi-parametric policy. This lets the method retain the variance-reduction benefits of locality while adding the generalization capacity of parametric prediction.

Smoothness-Constrained Policy Learning: A second line of work improves stability by adding explicit smoothness constraints, making it the closest conceptual comparison to our Laplacian-smoothing analysis. L2C2 Kobayashi (2022) studies model-free RL with local Lipschitz continuity constraints to improve smoothness and noise robustness, and Asadi et al. (2018) propose related Lipschitz constraints for model-based RL. CCIL Ke et al. (2024a) adapts this idea to imitation learning by using a Lipschitz-constrained dynamics model to generate corrective labels, and similar smoothness objectives have been scaled to humanoid controllers Chen et al. (2024). DARP targets the same quantitative behavior—lower local variance and improved rollout stability—but obtains it through retrieval aggregation inside the architecture while keeping the training objective in the standard imitation-learning form.

In-Context Learning Methods: Retrieval has also been studied as in-context imitation learning, where the comparison set is supplied directly to a sequence model. REGENT Sridhar et al. (2025) conditions a causal transformer on retrieved states, actions, and rewards, while DPT Lee et al. (2023) pretrains transformers to predict actions from query states and in-context datasets. Other in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). These methods emphasize fast adaptation to new tasks and environments. In contrast, DARP is evaluated as a standard imitation-learning policy under matched demonstrations, with the empirical focus on margins over BC, consistency across representations, and the architectural sources of stability.

5 CONCLUSION

We introduced **Difference-Aware Retrieval Policies for Imitation Learning (DARP)** (DARP), a nearest-neighbor-based policy that reframes imitation learning around quantitative comparisons between a query state and retrieved expert neighbors. Instead of learning only a direct state-to-action map, DARP predicts from relative differences $(s_i^*, a_i^*, s_i^* - s_q)$ and aggregates the resulting neighbor-level actions. Theoretically, this architecture implements implicit Laplacian smoothing, giving a variance-reduction mechanism without adding a tuned regularizer. Empirically, the evidence is consistent across the main comparisons: DARP outperforms standard BC in low-dimensional continuous-control and manipulation settings, preserves favorable margins under visual features and richer action distributions, and yields 15–46% improvements across the tested scenarios. The ablations further localize the gains to the difference-vector representation and permutation-invariant aggregation, showing that the practical benefit comes from how retrieved evidence is compared and combined rather than from retrieval alone.

6 REPRODUCIBILITY STATEMENT

Supplementary source code, policy and environment configuration files, MuJoCo experiment data, scripts for generating MimicGen expert demonstrations for Robosuite tasks, and modality-transformation code will be released upon acceptance. The reported comparisons are expected to reproduce exactly on NVIDIA L40 and L40s GPUs, except for transformer-based results (REGENT and Set Transformer), whose non-deterministic training may lead to slight numerical deviations from the table values while preserving the same evaluation setup.

REFERENCES

- Kavosh Asadi, Dipendra Misra, and Michael Littman. Lipschitz continuity in model-based reinforcement learning. In *International Conference on Machine Learning*. PMLR, PMLR, 2018.
- Mikhail Belkin and Partha Niyogi. Towards a theoretical foundation for laplacian-based manifold methods. *Journal of Computer and System Sciences*, 74(8):1289–1308, 2008.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0 : A vision-language-action flow model for general robot control, 2024. URL <https://arxiv.org/abs/2410.24164>.

-
- Zixuan Chen, Xialin He, Yen-Jen Wang, Qiayuan Liao, Yanjie Ze, Zhongyu Li, S. Shankar Sastry, Jiajun Wu, Koushil Sreenath, Saurabh Gupta, and Xue Bin Peng. Learning smooth humanoid locomotion through lipschitz-constrained policies. *arxiv preprint arXiv:2410.11825*, 2024.
- Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2024.
- Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997. doi: 10.1090/cbms/092.
- Michael Jae-Yoon Chung, Maxwell Forbes, Maya Cakmak, and Rajesh PN Rao. Accelerating imitation learning through crowdsourcing. In *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4777–4784. IEEE, 2014.
- Norman Di Palo and Edward Johns. Keypoint action tokens enable in-context imitation learning in robotics. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
- Maximilian Du, Suraj Nair, Dorsa Sadigh, and Chelsea Finn. Behavior retrieval: Few-shot imitation learning by querying unlabeled datasets. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4rl: Datasets for deep data-driven reinforcement learning, 2020.
- Letian Fu, Huang Huang, Gaurav Datta, Lawrence Yunliang Chen, William Chung-Ho Panitch, Fangchen Liu, Hui Li, and Ken Goldberg. In-context imitation learning via next-token prediction. *arXiv preprint arXiv:2408.15980*, 2024.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha Srinivasa, and Abhishek Gupta. Ccil: Continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha S. Srinivasa, and Abhishek Gupta. CCIL: continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024b. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Taisuke Kobayashi. L2c2: Locally lipschitz continuous constraint towards stable and smooth reinforcement learning. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4032–4039. IEEE, IEEE, October 2022.
- Jonathan Lee et al. Supervised pretraining can learn in-context reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 43057–43083, 2023.
- Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosior, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, pp. 3744–3753, 2019.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems, 2020. URL <https://arxiv.org/abs/2005.01643>.
- Li-Heng Lin, Yuchen Cui, Amber Xie, Tianyu Hua, and Dorsa Sadigh. Flowretrieval: Flow-guided data retrieval for few-shot imitation learning. In Pulkrit Agrawal, Oliver Kroemer, and Wolfram Burgard (eds.), *Conference on Robot Learning, 6-9 November 2024, Munich, Germany*, volume 270 of *Proceedings of Machine Learning Research*, pp. 4084–4099. PMLR, 2024. URL <https://proceedings.mlr.press/v270/lin25a.html>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR) 2023*, 2023. URL <https://arxiv.org/abs/2210.02747>. Originally appeared as arXiv preprint arXiv:2210.02747.

-
- Jiafei Lyu, Xiaoteng Ma, Le Wan, Runze Liu, Xiu Li, , and Zongqing Lu. Seabo: A simple search-based method for offline imitation learning. In *International Conference on Learning Representations*, 2024.
- Ajay Mandlekar, Soroush Nasiriany, Bowen Wen, Ireteyio Akinola, Yashraj Narang, Linxi Fan, Yuke Zhu, and Dieter Fox. Mimicgen: A data generation system for scalable robot learning using human demonstrations. In *7th Annual Conference on Robot Learning*, 2023.
- Elman Mansimov and Kyunghyun Cho. Simple nearest neighbor policy method for continuous control tasks. In *International Conference on Learning Representations (ICLR) 2018*, 2018. Under review as a conference paper at ICLR 2018.
- Marius Memmel, Jacob Berg, Bingqing Chen, Abhishek Gupta, and Jonathan Francis. Strap: Robot sub-trajectory retrieval for augmented policy learning. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3M: A universal visual representation for robot manipulation. In Karen Liu, Dana Kulic, and Jeffrey Ichnowski (eds.), *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, volume 205 of *Proceedings of Machine Learning Research*, pp. 892–909. PMLR, 2022. URL <https://proceedings.mlr.press/v205/nair23a.html>.
- Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. Robocasa: Large-scale simulation of everyday tasks for generalist robots. In *Robotics: Science and Systems (RSS)*, 2024.
- Jyothish Pari, Nur Muhammad Shafiullah, Sridhar Pandian Arunachalam, and Lerrel Pinto. The surprising effectiveness of representation learning for visual imitation. In *Robotics: Science and Systems (RSS)*. Robotics: Science and Systems Foundation, June 2022. doi: 10.15607/RSS.2022.XVIII-052.
- Emmanuel Pignat and Sylvain Calinon. Bayesian gaussian mixture model for robotic policy imitation. *IEEE Robotics and Automation Letters*, 2019. URL <https://arxiv.org/pdf/1904.10716.pdf>. Preprint version.
- Dean Pomerleau. Efficient training of artificial neural networks for autonomous navigation. *Neural Comput.*, 3(1):88–97, 1991. doi: 10.1162/NECO.1991.3.1.88. URL <https://doi.org/10.1162/neco.1991.3.1.88>.
- Stéphane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pp. 661–668. JMLR Workshop and Conference Proceedings, 2010.
- Stéphane Ross, Geoffrey J. Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In Geoffrey J. Gordon, David B. Dunson, and Miroslav Dudík (eds.), *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, AISTATS 2011, Fort Lauderdale, USA, April 11-13, 2011*, volume 15 of *JMLR Proceedings*, pp. 627–635. JMLR.org, 2011. URL <http://proceedings.mlr.press/v15/ross11a/ross11a.pdf>.
- Steven L. Salzberg and David W. Aha. Learning to catch: Applying nearest neighbor algorithms to dynamic control tasks. In P. Cheeseman and R. W. Oldford (eds.), *Selecting Models from Data: Artificial Intelligence and Statistics IV*, volume 89 of *Lecture Notes in Statistics*, pp. 321–328. Springer, 1994. doi: 10.1007/978-1-4612-2660-4_33. URL https://link.springer.com/chapter/10.1007/978-1-4612-2660-4_33.
- Jonathan C. Spencer, Sanjiban Choudhury, Arun Venkatraman, Brian D. Ziebart, and J. Andrew Bagnell. Feedback in imitation learning: The three regimes of covariate shift. *CoRR*, abs/2102.02872, 2021. URL <https://arxiv.org/abs/2102.02872>.
- Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, and Insup Lee. REGENT: A retrieval-augmented generalist agent that can act in-context in new environments. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. Oral presentation.

-
- Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pp. 5026–5033. IEEE, 2012.
- Arun Venkatraman, Martial Hebert, and J. Andrew Bagnell. Improving multi-step prediction of learned time series models. In Blai Bonet and Sven Koenig (eds.), *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, January 25-30, 2015, Austin, Texas, USA*, pp. 3024–3030. AAAI Press, 2015. doi: 10.1609/AAAI.V29I1.9590. URL <https://doi.org/10.1609/aaai.v29i1.9590>.
- Jan Ole von Hartz, Adrian Röfer, Joschka Boedecker, and Abhinav Valada. The unreasonable effectiveness of discrete-time gaussian process mixtures for robot policy learning. *arXiv preprint arXiv:2505.03296*, 2025.
- Vitalis Vosylius and Edward Johns. Instant policy: In-context imitation learning via graph diffusion. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan R Salakhutdinov, and Alexander J Smola. Deep sets. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (eds.), *Advances in Neural Information Processing Systems 30*, pp. 3391–3401. Curran Associates, Inc., 2017. URL <http://papers.nips.cc/paper/6931-deep-sets.pdf>.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- Dengyong Zhou, Olivier Bousquet, Thomas Navin Lal, Jason Weston, and Bernhard Schölkopf. Learning with local and global consistency. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 16, 2003.
- Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Soroush Nasiriany, Yifeng Zhu, and Kevin Lin. robosuite: A modular simulation framework and benchmark for robot learning. In *arXiv preprint arXiv:2009.12293*, 2020.

A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

To prove Theorem 1, we first start with a well-known result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2} (\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \rightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Thus, under the stated assumptions, MRIL has strictly stronger generalization and stability guarantees than BC under the same expert-data setting.

Proof. We prove each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top Lf,$$

where $L = D - W$ is the graph Laplacian, and W is the k -NN affinity matrix. By Lemma 1, this equals the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top Lf$. In statistical learning theory, adding a Tikhonov penalty strictly reduces the variance of the estimator compared to the unregularized solution while keeping the bias term controlled. Thus MRIL enjoys smaller estimator variance than vanilla BC, which uses no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C \|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Therefore minimizers of $\mathcal{L}_{\text{MRIL}}$ have uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, minimizers of vanilla BC have no such constraint: any oscillatory interpolant that matches the training data exactly yields the same supervised risk, so arbitrarily large Lipschitz constants are possible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is only minimized on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a(O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly slower than in vanilla BC. In particular, error growth is sublinear in the rollout horizon when h is small, whereas it is linear for vanilla BC.

Conclusion. Claims (i)–(iii) establish that MRIL yields lower variance, uniform smoothness control, and sublinear rollout error accumulation compared to vanilla BC, completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This choice is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and ensures that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. In practice, we found this default to be stable across tasks, though other kernels (e.g., uniform k -NN or exponential decay) yield qualitatively similar results.

A.1.2 PROOF OF THEOREM 2

We begin with the required Lemmas establishing the spectral form of explicit Laplacian regularization and neighbor aggregation.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U \Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = U c$, $a^* = U b$ in this basis. The objective becomes

$$\|U c - U b\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = S f$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2}AD^{-1/2}$ and $S = D^{-1}A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1}L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $Su_j = (1 - \mu_j)u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

This spectral comparison frames DARP as *locally adaptive implicit regularization*. Instead of selecting a global weight λ , the neighbor aggregation step uses the graph structure itself to determine how strongly nearby predictions are smoothed. The effective strength therefore varies with local degree and neighborhood geometry, matching the density of the expert demonstrations. In spectral terms, the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$ suppresses high-frequency modes more aggressively than a fixed explicit- λ filter, as illustrated in Figure 8.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

The distance function $d(s_q, s_i^*)$ determines which empirical evidence enters each DARP prediction, so retrieval quality directly affects the reported margins. Simple Euclidean distance between states can work, but our experiments use a history-aware distance because the data are trajectories.

Let the query trajectory be $S_q = (\dots, s_{q,-1}, s_{q,0})$, with $s_{q,0}$ as the current query state s_q . To compute

$d(s_q, s_i^*)$ for an expert state s_i^* , we identify its source trajectory S_j^* and time index t , so s_i^* can be written as $s_{j,t}^*$. For a lookback parameter ℓ , the unweighted history distance is:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

This metric sums Euclidean discrepancies over the current state and the preceding $\ell - 1$ states, using valid indices from the query and source trajectories. Because recent observations are typically more informative for selecting the relevant neighbor set, we also use an exponentially decayed variant with rate $\gamma \geq 0$:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how Stack success changes as k , lookback length, and decay rate are swept, making retrieval sensitivity explicit.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We evaluate multimodal action modeling on Push-T (Chi et al., 2024), visualized in Figure 10. In this condition, DARP uses a Set Transformer head to predict Gaussian Mixture Model parameters. The resulting score is 70 ± 8 compared with 48 ± 8 for BC, a 22-point improvement over 100 trials (Table 4). This comparison shows that the retrieval-and-difference architecture remains favorable when the output model must represent multiple plausible actions.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A central risk for neighbor-based methods is discontinuity: Euclidean-near states may require opposing actions. We therefore use a modified D4RL Umaze stress test (Figure 11) to compare success under this failure mode.

In this deliberately discontinuous setting, DARP achieves 57% success versus 25% for BC (Table 5). The 32-point margin indicates that difference vectors and permutation-invariant aggregation help filter appropriate from inappropriate neighbors even when spatial proximity alone is ambiguous.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To measure robustness to accumulated BC error, we roll out a BC policy known to fail and, every k steps, fork the environment and deploy DARP from the copied state. Figure 12 shows that DARP often recovers high reward even from states late in the BC trajectory. This comparison supports the claim that DARP is more robust to the slightly OOD states produced by compounding BC errors.

A.2.5 COMPARISON WITH CCIL

Table 6 compares DARP with CCIL, an explicit-smoothness baseline, using the percent improvement over each method’s BC reference. DARP is higher on three of four environments and ties on Walker. The largest relative separation is on HalfCheetah, where DARP reports a 418.7% improvement over BC compared with 5.4% for CCIL.

A.2.6 DISTANCE METRIC SENSITIVITY

Although the main experiments use Euclidean distance for neighbor retrieval, high-dimensional embeddings such as R3M also motivate cosine similarity. Table 7 shows that cosine similarity reduces DARP from 75% to 70% success on Robosuite Stack, while both retrieval metrics remain well above the 47% BC baseline. The metric choice therefore changes the margin by 5 points but does not remove the advantage over BC.

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC compared with CCIL.** On MuJoCo benchmarks, DARP matches or exceeds CCIL’s percent improvement over BC in every task, with especially large margins on Ant and HalfCheetah.

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance metric sensitivity.** On Robosuite Stack with R3M features, DARP reaches 75% success with Euclidean distance and 70% with cosine similarity, compared with 47% for BC.

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP combined with diffusion policy.** Success rates over 50 trials on state-based Robosuite Stack, Threading, and Peg Insertion show that DARP improves both MLP and diffusion backbones. This experiment uses fewer demonstrations than Table 2, so the BC and DARP values are not directly matched to that table.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Diffusion policy (Chi et al., 2024) is a strong imitation-learning model class for manipulation, so Table 8 tests whether DARP is complementary to it. With a diffusion backbone, DARP improves over DARP with an MLP backbone by 20, 34, and 26 success-rate points on Stack, Threading, and Peg Insertion, respectively. Relative to BC (MLP), the total gains are 48, 60, and 38 points, showing that the retrieval architecture can add to a stronger action model rather than only improving a simple MLP.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP with an MLP backbone versus diffusion policy.** In this comparison, diffusion is $\approx 10\times$ slower per training epoch and $\approx 36\times$ slower per inference step than DARP (MLP).

Table 9 quantifies the efficiency tradeoff: DARP with an MLP backbone is much faster than standard diffusion, especially at inference time.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** On Robosuite Stack with human-collected demonstrations, DARP reaches 60% success versus 45% for BC.

Most MuJoCo and Robosuite demonstrations in the main experiments come from an optimal reinforcement-learning policy, so Table 10 checks whether the margin persists with human demonstrations. On Robosuite Stack, DARP reaches 60% success compared with 45% for BC, a 15-point improvement under human-collected data.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Validation-selected retrieval settings.** When epochs and retrieval hyperparameters are selected by validation loss, DARP remains higher than BC on Hopper (805.69 vs 507.49) and Stack (0.31 vs 0.12).

Figure 9 shows that poor retrieval choices, such as very small neighborhoods, can make DARP worse than BC. Table 11 shows that validation-loss selection identifies settings where DARP preserves a clear advantage over BC on both tested environments.

A.3 PSEUDOCODE

Algorithm 1 summarizes the training loop used in the comparisons: retrieve neighbors for each query, form difference-conditioned predictions, aggregate them, and optimize the standard imitation loss.

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Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \arg \min -k_j d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime comparison across environments.** Training time per epoch and test time per step for BC and DARP with varying k on Hopper and Stack. The table reports the computational cost associated with increasing neighborhood size.

Table 12 reports the training and inference overhead of DARP as k varies. The measured cost grows sub-linearly with neighborhood size in these settings, and inference remains practical for real-time control. For example, on Robosuite Stack with $k = 500$, inference takes approximately 0.00437 seconds per step on our hardware, corresponding to more than 230 Hz.

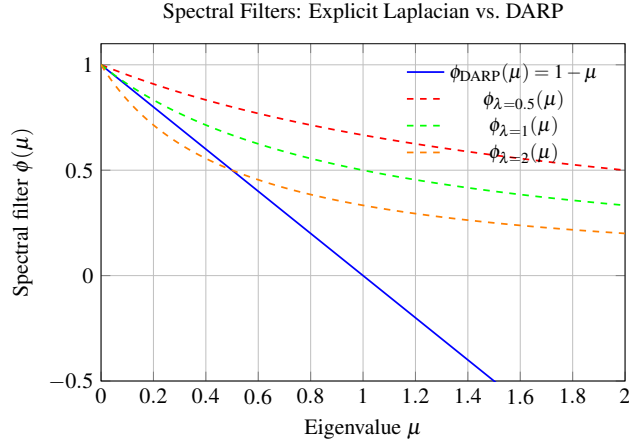


Figure 8: DARP applies a sharper low-pass spectral filter than the explicit Laplacian regularizer across high-frequency modes.

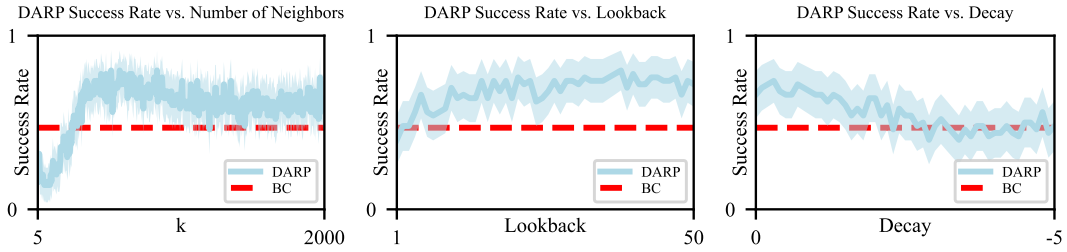


Figure 9: **Retrieval-hyperparameter sweeps on Robosuite Stack.** Success rates over 50 trials with 95% confidence intervals show three empirical patterns: (left) very small neighborhoods underperform, performance peaks around 500 neighbors, and very large k plateaus just above BC; (center) adding trajectory history generally improves DARP except when little or no history is used; (right) excessive decay on older observations makes retrieval behave like short-history retrieval and can fall below BC.

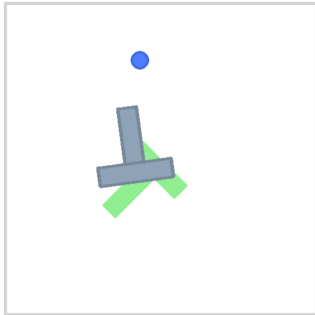


Figure 10: **Push-T environment.** The policy controls the blue circle to push the T-shaped block, a setting with multimodal action choices.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T results.** Over 100 trials, DARP scores 70 ± 8 versus 48 ± 8 for BC, a 22-point improvement.

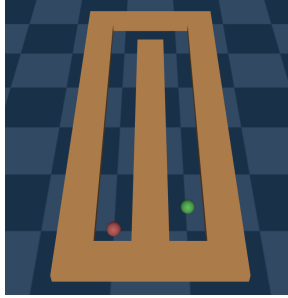


Figure 11: **Long maze environment.** The robot must move a force-actuated ball from the green start to the red destination in a discontinuous navigation setting.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Over 100 trials, DARP reaches 57% success versus 25% for BC.

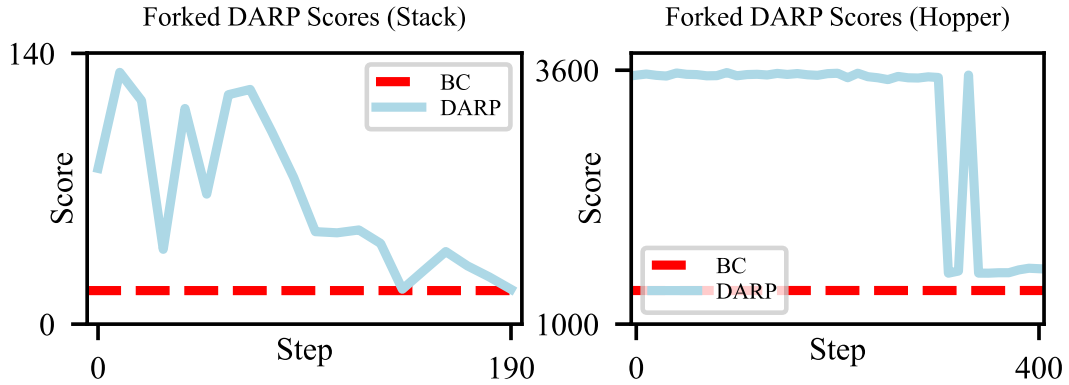


Figure 12: **Forked-rollout robustness comparison.** On Robosuite Stack and MuJoCo Hopper, we roll out a failing BC policy and fork the environment every $k = 10$ steps to start DARP from the same state. DARP maintains high scores from many BC-visited states and is only substantially limited about halfway through the Stack rollout and about 80% through the Hopper rollout.